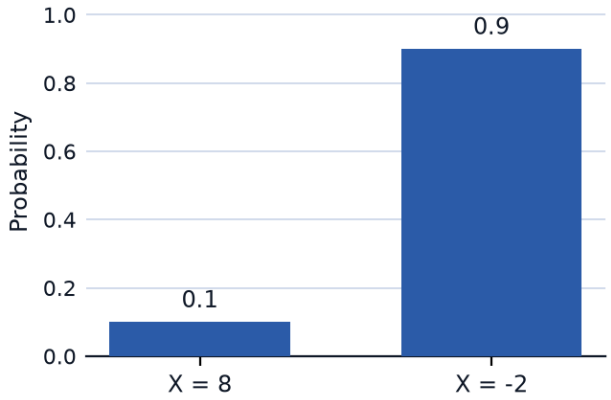
Discrete Random Variables & Binomial Distributions

ClipSAT Printable Practice Suite (clipsat.org) • Topic 3.02

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

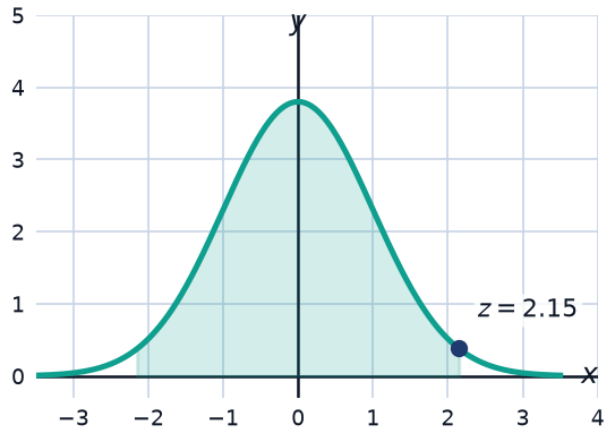
1 [Foundational] • Target: Probability distributions and expected value

A game costs 2 dollars to play. You win 10 dollars with probability 0.1, and win nothing otherwise. Let X be your net profit. Find the probability distribution of X and its expected value.



2 [Foundational] • Target: Probability distributions and expected value

Find the standard deviation of the profit X from the previous problem, to two decimal places.



3 [Foundational] • Target: Binomial setting and conditions

State the four conditions required for a random variable to follow a binomial distribution (sometimes remembered as BINS: Binary, Independent, Number fixed, Same probability).

4 [Foundational] • Target: Binomial setting and conditions

A survey randomly selects 20 people (without replacement) from a small town of 50 people and records how many support a proposal. Explain why this technically violates the independence condition, and why it is often still treated as approximately binomial.

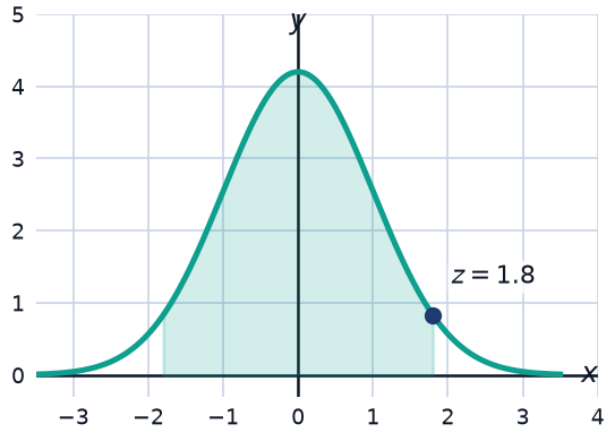
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5 [Foundational] • Target: Binomial probability formula

A basketball player makes 70% of free throws. Find the probability she makes exactly 4 out of 6 attempts, using $P(X = k) = \binom{n}{k} p^k (1-p)^{n-k}$.

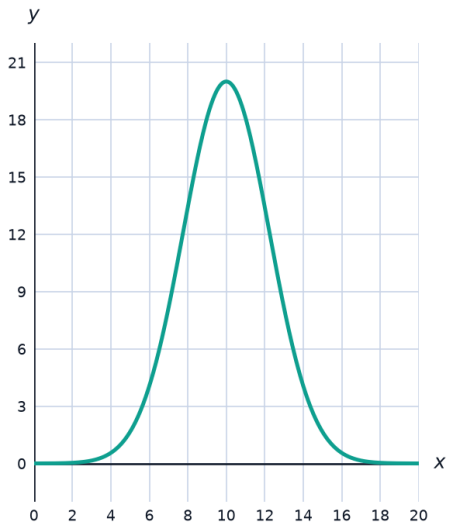
6 [Foundational] • Target: Binomial probability formula

Using the same free-throw shooter, find the mean and standard deviation of the number of makes out of 6 attempts, using $\mu = np$ and $\sigma = \sqrt{np(1-p)}$.



7 [Foundational] • Target: Visualizing a binomial distribution

The graph shows a smoothed approximation of the binomial distribution for $n = 20$ trials with $p = 0.5$, plotted as a continuous curve for visual comparison to a normal shape.



a Using $\mu = np$ and $\sigma = \sqrt{np(1-p)}$, verify that this binomial distribution has mean 10 and standard deviation approximately 2.24, matching the curve shown.

b Explain why a binomial distribution with $n = 20$ and $p = 0.5$ looks approximately bell-shaped, referencing the conditions for a normal approximation.

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Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 **[Intermediate]** • **Target: Geometric distribution (brief introduction)**

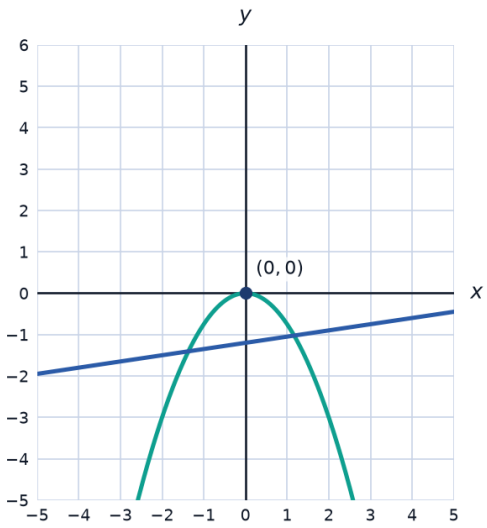
A basketball player makes free throws with probability 0.7 each attempt, independently. Find the probability that her first miss occurs on the 3rd attempt (i.e. she makes the first 2, then misses the 3rd).

9 **[Intermediate]** • **Target: Applying the binomial distribution in context**

A multiple-choice quiz has 10 questions, each with 4 choices, and a student guesses randomly on every question. Find the probability the student gets at least 8 questions correct, given that $P(X = 8) \approx 0.0004$, $P(X = 9) \approx 0.00003$, and $P(X = 10) \approx 0.000001$.

10 **[Intermediate]** • **Target: Binomial probability of at least one success**

A manufacturing process produces defective items independently with probability 0.05. In a batch of 20 items, find the probability that at least one item is defective, using the complement of "zero defective."



11 **[Intermediate]** • **Target: Binomial probability of at least one success**

Using the mean and standard deviation formulas $\mu = np$ and $\sigma = \sqrt{np(1-p)}$, find the mean and standard deviation of the number of defective items in a batch of 20, with $p = 0.05$.

12 **[Intermediate]** • **Target: Comparing binomial scenarios**

Explain why a binomial model would NOT be appropriate for counting the number of aces dealt when drawing 5 cards without replacement from a standard 52-card deck, and name the distribution that would be more appropriate instead.

13 **[Intermediate]** • **Target: Geometric distribution (brief introduction)**

A discrete random variable X has probability distribution: $P(X = 0) = 0.1$, $P(X = 1) = 0.3$, $P(X = 2) = 0.4$, $P(X = 3) = 0.2$. Calculate: (a) $E(X)$, (b) $E(X^2)$, and (c) $\text{Var}(X)$ and σ_X.

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14 [Intermediate] • Target: Applying the binomial distribution in context

Independent random variables X and Y have $\mu_X = 50$, $\sigma_X = 6$ and $\mu_Y = 30$, $\sigma_Y = 8$. Define $W = 3X - 2Y + 10$. Find the mean μ_W and standard deviation σ_W.

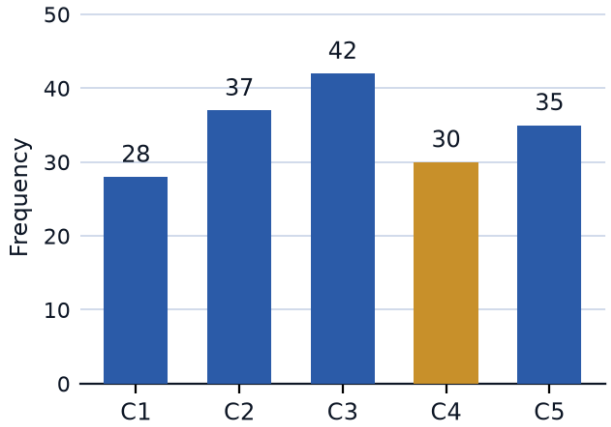
Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 [Challenging] • Target: Binomial probability of at least one success • Cross-Lesson: 1.01 Measures of Center & Spread

Level 1 · FoundationsThe temperatures (in °C) during a science experiment have median $M = 22.0^\circ\text{C}$ and IQR $= 4.5^\circ\text{C}$. Convert the temperatures to Fahrenheit using the formula $F = 1.8C + 32$. What are the median and IQR of the temperatures in °F?

16 [Challenging] • Target: Comparing binomial scenarios • Cross-Lesson: 1.02 Z-Scores & Linear Transformations

The number of accidents on a highway follows a Poisson distribution with mean 1.6 per week. (a) Find the probability of at least 3 accidents in a 2-week period. (b) Given that at least one accident occurs in a week, find the probability that there are exactly 2 accidents in that week.



17 [Challenging] • Target: Probability distributions and expected value • Cross-Lesson: 1.03 Describing Distributions & Linear Regression

A discrete random variable X follows a binomial distribution $X \sim B(15, 0.35)$. (a) Find $P(X = 5)$. (b) Find $P(X \geq 4)$. (c) Calculate the mean $E(X)$ and standard deviation σ of X.

18 [Challenging] • Target: Binomial setting and conditions • Cross-Lesson: 2.01 Sampling Methods & Bias

Practice 4.1: A multiple-choice examination consists of 20 independent questions, each having 4 options of which only one is correct. A student guesses randomly on all questions. Let X be the number of correct answers. (a) State the distribution of X including its parameters. (b) Find the probability that the student passes by scoring at least 10 correct answers. (c) Calculate the mean and standard deviation of X.

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19 [Challenging] • Target: Binomial probability formula • Cross-Lesson: 2.02 Experimental Design & Scope of Inference

Cars arrive at a toll booth at an average rate of 3 per minute. Find the probability that (a) exactly 5 arrive in one minute, (b) at least 2 arrive in 30 seconds.

20 [Challenging] • Target: Visualizing a binomial distribution • Cross-Lesson: 3.01 Probability Rules & Conditional Probability

A diagnostic medical screening test is 95% sensitive (positive if disease present) and 90% specific (negative if disease absent). In a population, 2% of people have the disease. Find the probability that a person who tests positive actually has the disease.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: Probability distributions and expected value

A binomial random variable has $n = 14$ trials with success probability $p = 0.7$. Find the mean $\mu = np$.

a. 4.2

b. 10.8

c. 8.8

d. 9.8

22 [Foundational] • Target: Binomial setting and conditions

A binomial random variable has $n = 15$ trials with success probability $p = 0.9$. Find the standard deviation $\sigma = \sqrt{np(1-p)}$, to two decimal places.

a. 1.35

b. 1.16

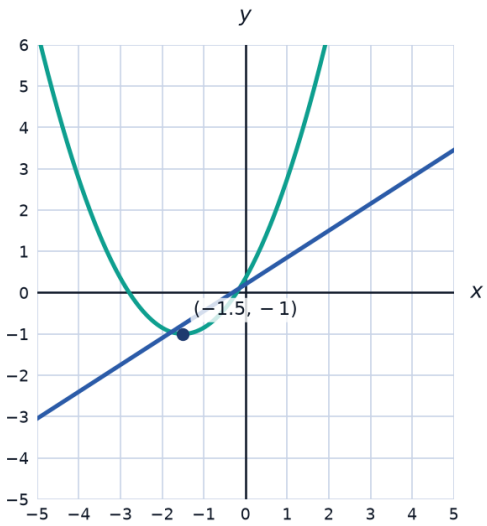
c. 0.86

d. 1.46

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23 [Foundational] • Target: Binomial probability formula

A manufacturing process produces defective items independently with probability 0.3. In a batch of 15 items, find the probability that at least one item is defective, using the complement of "zero defective," to three decimal places.



a. 1.045

b. 0.995

c. 0.005

d. 0.945

24 [Foundational] • Target: Visualizing a binomial distribution

A process succeeds independently with probability $p = 0.5$ on each of $n = 10$ trials. Find the probability of exactly 9 successes, using $P(X = k) = \binom{n}{k} p^k (1 - p)^{n - k}$, to three decimal places.

a. 0.001

b. 0.020

c. 0.060

d. 0.010

25 [Foundational] • Target: Geometric distribution (brief introduction)

A game costs 5 dollars to play. You win 15 dollars with probability 0.9, and win nothing otherwise. Letting X be your net profit, find $E(X)$.

a. $-\frac{7}{2}$

b. $\frac{15}{2}$

c. $\frac{17}{2}$

d. $\frac{19}{2}$

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26 [Foundational] • Target: Applying the binomial distribution in context

Binomial distribution requires:

a. Continuous outcomes

b. Fixed n, two outcomes, constant p, independence

c. Normal data

d. Dependent trials

27 [Foundational] • Target: Binomial probability of at least one success

A binomial random variable has $n = 6$ trials and success probability $p = 0.25$. Find $P(X = 6)$.

a. 0.0044

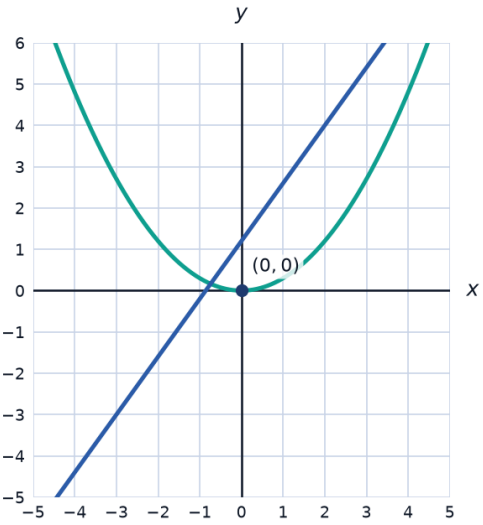
b. 0.0002

c. 0.1318

d. 0.033

28 [Foundational] • Target: Comparing binomial scenarios

A geometric random variable counts trials until the first success, with success probability $p = 0.29$ each trial. Find $P(X = 6)$, the probability the first success occurs on trial 6.



a. 0.0737

b. 0.1462

c. 0.0523

d. 0.0264

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Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 **[Intermediate]** • **Target: Probability distributions and expected value**

A binomial random variable has $n = 30$ and $p = 0.24$. Find its standard deviation $\sigma = \sqrt{np(1-p)}$.

a. 2.339

b. 0.339

c. 1.839

d. 1.339

30 **[Intermediate]** • **Target: Binomial setting and conditions**

Normal distribution is symmetric about:

a. Standard deviation

b. Median only

c. The mean

d. Zero

31 **[Intermediate]** • **Target: Binomial probability formula**

The Central Limit Theorem states that sample means are approximately Normal when:

a. Population is Normal

b. n is large (n≥30)

c. σ is known

d. p<0.5

32 **[Intermediate]** • **Target: Visualizing a binomial distribution**

A Normal random variable has $\mu = 189$ and $\sigma = 8$. Find $P(173 < X < 213)$.

a. 0.8759

b. 0.9759

c. 0.7759

d. 0.6759

33 **[Intermediate]** • **Target: Geometric distribution (brief introduction)**

Events A and B satisfy $P(A) = 0.55$, $P(B) = 0.12$, and $P(A \cap B) = 0.0$. Find $P(A \cup B)$.

a. 0.57

b. 0.67

c. 0.82

d. 0.62

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34 [Intermediate] • Target: Applying the binomial distribution in context

P(A and B) if A,B independent:

a. $P(A) + P(B)$

b. $P(A) \cdot P(B)$

c. $P(A|B)$

d. $P(A) + P(B) - P(A \cup B)$

35 [Intermediate] • Target: Binomial probability of at least one success

In a standard Normal, what % of values fall within 1σ?

a. ~68%

b. ~95%

c. ~99.7%

d. ~50%

36 [Intermediate] • Target: Comparing binomial scenarios

A random variable X follows a binomial distribution $B(n, p)$. The number of trials is $n = 10$ and the probability of success in a single trial is p. If the mean of X is 4, find the variance of X.

a. 2.4

b. 3.6

c. 6.4

d. 7.2

37 [Intermediate] • Target: Probability distributions and expected value

A researcher is designing a survey to understand student preferences for extracurricular activities at a large university. The university has 4 distinct colleges: Arts & Sciences, Engineering, Business, and Health Sciences. The researcher decides to use stratified random sampling. If each college represents one stratum, and the researcher wants to survey 200 students in total, with the number of students surveyed from each college being proportional to the college's enrollment, which of the following represents a valid way to determine the sample size for each college?

a. A simple random sample of 200 students from the entire university population.

b. A systematic sample of 200 students from an alphabetized list of all students.

c. A convenience sample of 200 students from the campus cafeteria.

d. Proportionally allocating the 200 students to each of the 4 colleges based on their enrollment figures.

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Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 **[Challenging] • Target: Binomial setting and conditions • Cross-Lesson: 2.01 Sampling Methods & Bias**

A histogram is used to display the distribution of the number of hours college students reported spending on social media per day. The bins for the histogram are as follows: 0-2 hours, 2-4 hours, 4-6 hours, 6-8 hours, and 8-10 hours. The heights of the bars are 5, 12, 15, 8, and 3, respectively. What is the total number of students represented by this histogram?

a. 15

b. 43

c. 50

d. 200

39 **[Challenging] • Target: Binomial probability formula • Cross-Lesson: 2.02 Experimental Design & Scope of Inference**

A distribution is displayed as a symmetric, bell-shaped histogram with a single peak in the middle. How should this shape be described?

a. Bimodal

b. Skewed left (negatively skewed)

c. Skewed right (positively skewed)

d. Approximately symmetric (unimodal)

40 **[Challenging] • Target: Visualizing a binomial distribution • Cross-Lesson: 3.01 Probability Rules & Conditional Probability**

A distribution is displayed as a histogram with two separate peaks. How should this shape be described?

a. Skewed right (positively skewed)

b. Approximately symmetric (unimodal)

c. Skewed left (negatively skewed)

d. Bimodal

41 **[Challenging] • Target: Geometric distribution (brief introduction) • Cross-Lesson: 3.03 The Normal Distribution**

A distribution is right-skewed when:

a. Mean < Median

b. Mean > Median

c. Mean = Median

d. Mode > Mean

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42 [Challenging] • Target: Applying the binomial distribution in context • Cross-Lesson: 3.04 Sampling Distributions & Central Limit Theorem

A wire of length 4 m is bent to form a rectangle with adjacent sides of lengths X m and $(2 - X)$ m, where X is a continuous random variable with probability density function $f(x) = \begin{cases} \frac{3}{8}x^2, & 0 \le x \le 2, \\ 0, & \text{otherwise.} \end{cases}$ Given that the area of the rectangle is at least $\frac{3}{4}$ m^2, find the probability that the length of the shorter side is at most $\frac{2}{3}$ m.

a. $\frac{127}{351}$

b. $\frac{74}{351}$

c. $\frac{37}{351}$

d. $\frac{127}{640}$

43 [Challenging] • Target: Binomial probability of at least one success • Cross-Lesson: 4.01 Confidence Intervals & Hypothesis Testing

The bar graph below shows the distribution of the number of siblings reported by the 25 students in a homeroom class. If 1 student is selected at random from the class, what is the probability that the selected student has at least 2 siblings?

a. $\frac{6}{25}$

b. $\frac{9}{25}$

c. $\frac{12}{25}$

d. $\frac{16}{25}$

44 [Challenging] • Target: Comparing binomial scenarios • Cross-Lesson: 4.02 Chi-Square Tests & Regression Inference

Which of the following conditions is necessary to establish a cause-and-effect relationship between two variables?

a. A large sample size greater than 10,000

b. Random selection of participants from the population

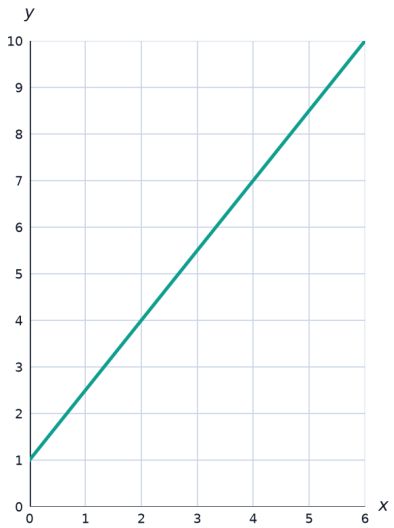
c. Random assignment of participants to experimental treatment groups

d. A correlation coefficient of exactly 1.0

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45 [Challenging] • Target: Probability distributions and expected value • Cross-Lesson: 1.01 Measures of Center & Spread

In $\hat{y} = a + bx$, the slope b represents:



a. The predicted value at x=0

b. Change in y per unit x

c. Correlation coefficient

d. Standard error

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